Question

The table below presents the adsorption data of $AuCl_4^-$ at different concentrations on the surface of a certain material under the same temperature, which can be used to evaluate the interactions between adsorbed species on the material surface.

c /(mmol/L)	q_e / (mmol $AuCl_4^-$ /g material)
0.114	5.706
0.205	6.935
0.310	7.595
0.405	7.921

Two commonly used models are the Langmuir model and the Freundlich model. In these models, the relationship between the equilibrium adsorption capacity q_e and the concentration c of the species to be adsorbed in the solution (i.e., the adsorption isotherm) is given as follows (where q_m represents the maximum adsorption capacity per unit mass of the material surface, K_L and K_f are the equilibrium constants of the adsorption process, and n is a related parameter):

Langmuir model:

$$q_e = q_m K_L c / (1 + K_L c)$$

Freundlich model:

$$q_e = K_f c^{1/n}$$

By analyzing the fitting results of different adsorption models, select the correct statement from the following options.

A. The interaction between Au species adsorbed on the material surface is negligible, and the equilibrium constant for the adsorption process is 8.

B. The interactions between adsorbed Au species on the material surface are negligible, and the equilibrium constant for the adsorption process is 14.

C. The interaction between Au species adsorbed on the material surface cannot be ignored, and the equilibrium constant for the adsorption process is 8.

D. The interaction between Au species adsorbed on the material surface cannot be neglected, and the equilibrium constant for the adsorption process is 14.

E. The interaction between Au species adsorbed on the material surface is negligible. According to the expression, the maximum adsorption capacity (in terms of Au) is calculated to be 1800 mg Au / g material.

F. The interaction between adsorbed Au species on the material surface is negligible, and the maximum adsorption capacity (calculated in terms of Au) derived from the expression is 1600 mg Au / g material.

G. The interaction between Au species adsorbed on the material surface cannot be neglected. The maximum adsorption capacity (calculated based on Au content) derived from the expression is 1800 mg Au / g material.

H. The interaction between Au species adsorbed on the material surface cannot be ignored. The maximum adsorption capacity (calculated in terms of Au) obtained from the expression is 1600 mg Au / g material.

Answer

Correct Answer: B

Detailed Explanation

First, the Langmuir model is discussed. By rearranging the adsorption isotherm equation, we obtain:

$$1/q_\mathrm{e} = 1/(q_\mathrm{m} K_\mathrm{L} c) + 1/q_\mathrm{m}$$

CHECKPOINT 1 PTS

$$1/q_\mathrm{e} = 1/(q_\mathrm{m} K_\mathrm{L} c) + 1/q_\mathrm{m}$$

Thus, by performing linear regression with $1/c$ as the independent variable and $1/q_\mathrm{e}$ as the dependent variable, we obtain:

$$1/q_\mathrm{e} = 0.0078(1/c) + 0.1066(R^2 = 0.99986)$$

CHECKPOINT 1 PTS

$$1/q_\mathrm{e} = 0.0078(1/c) + 0.1066(R^2 = 0.99986)$$

Next, the Freundlich model is discussed. By rearranging the adsorption isotherm equation, we obtain:

$$\ln q_\mathrm{e} = \ln K_f + (1/n)\ln c$$

CHECKPOINT 1 PTS

$\ln q_e = \ln K_f + (1/n)\ln c$

By performing linear regression with $\ln c$ as the independent variable and $\ln q_e$ as the dependent variable, we obtain:

$$\ln q_e = 2.32 + 0.26\ln c (R^2 = 0.98927)$$

CHECKPOINT 1 PTS

$\ln q_e = 2.32 + 0.26\ln c (R^2 = 0.98927)$

Therefore, based on the fitted R^2 values, the adsorption of $AuCl_4^-$ is more consistent with the Langmuir model, indicating that the adsorbed Au species form a monolayer with negligible interactions. Thus, options C, D, G, and H are incorrect.

CHECKPOINT 1 PTS

The interactions between adsorbed Au species on the material surface are negligible

The corresponding saturation adsorption capacity can be calculated from the intercept:

$$q_m = 1/0.1066 = 9.381 \text{mmol } AuCl_4^- \text{/g material}$$

Converting this to mass yields the maximum adsorption capacity (in terms of Au) as 1848 mg Au / g material. Thus, option B is correct.

CHECKPOINT 1 PTS

The saturation adsorption capacity can be calculated from the intercept

CHECKPOINT 1 PTS

The maximum adsorption capacity (in terms of Au) is 1848 mg Au / g material

The equilibrium constant of the adsorption process can be calculated from the ratio of the intercept to the slope:

$K_{\mathrm{L}} = 0.1066/0.0078 = 14$

CHECKPOINT 1 PTS

The saturation adsorption capacity can be calculated from the ratio of the intercept to the slope

CHECKPOINT 1 PTS

$K_{\mathrm{L}} = 0.1066/0.0078 = 14$

Therefore, the correct answer is B.